

# Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

## Progress (as of r0009, 2026-0806)

| # | Quantity                         | Done             | Remaining | Of  |
|---|----------------------------------|------------------|-----------|-----|
| 1 | Definitions to define            | <b>4</b>         | 9         | ≈13 |
| 2 | <b>Numbered</b> results complete | <b>1</b> (Thm 7) | 27        | 28  |
| 3 | Mathlib foundations reused       | 12               | —         | 12  |
| 4 | sorry in the development         | —                | <b>0</b>  | —   |

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30) and so understates what is proved. The development contains 55 theorems, 15 definitions and 12 instances; of the theorems, these **eight are claims the paper makes in prose** rather than as numbered results, and all eight are formally verified:

| # | Paper claim                                                                        | Where                 | Lean                                                                                           |
|---|------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------|
| 1 | “the compact elements [of $P \rightarrow N$ ] are just the finite subsets of $N$ ” | p. 9                  | <code>isCompactElement_iff_finite</code>                                                       |
| 2 | “ $P \rightarrow N$ ... is a domain”                                               | p. 9                  | <code>instance : Domain (Set X)</code>                                                         |
| 3 | “ $D \rightarrow E$ is a ... cpo”                                                  | Thm 7 proof           | <code>instance : CompletePartialOrder (ScottHom <math>\alpha</math> <math>\beta</math>)</code> |
| 4 | “... bounded complete ... whenever $E$ is”                                         | Thm 7 proof           | <code>instance : BoundedComplete (ScottHom <math>\alpha</math> <math>\beta</math>)</code>      |
| 5 | “step(s) ... is continuous”                                                        | Thm 7 proof           | <code>scottContinuous_stepFun</code>                                                           |
| 6 | “... and compact in the ordering on $D \rightarrow E$ ”                            | Thm 7 proof           | <code>isCompactElement_step</code>                                                             |
| 7 | “they form a basis for $D \rightarrow E$ ”                                         | Thm 7 proof           | <code>instance : IsAlgebraic (ScottHom <math>\alpha</math> <math>\beta</math>)</code>          |
| 8 | every compact function is a <i>finite</i> join of step functions                   | Thm 7 proof, implicit | <code>exists_finite_isLUB_of_isCo</code>                                                       |

Six of those eight are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of  $D$  is never used.  $D$  need only be a domain. The function space is a cpo for any preordered  $D$ , algebraic when  $D$  and  $E$  are, bounded complete because  $E$  is, and countably based because  $D$  and  $E$  are.

The remaining ~47 theorems are supporting API: the  $\ll$  calculus, the `compactsBelow` machinery, the pointwise order and suprema on  $D \rightarrow E$ , and the step-function adjunction. The paper assumes or elides all of it.

**4 of ≈13 definitions are formally verified and building**, in 6 modules, 985 lines, 0 sorry, 0 warnings:

| # | Round | Module                                  | Contents                                                                                                                                                                                     |
|---|-------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | r0003 | <code>ScottDomains/WayBelow.lean</code> | way-below $\ll$ at <code>[Preorder <math>\alpha</math>]</code> , 7 theorems; <code>x <math>\ll</math> x <math>\leftrightarrow</math> IsCompactElement x</code> holds by <code>Iff.rfl</code> |

| # | Round       | Module                                       | Contents                                                                                                                                                                                                                   |
|---|-------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | r0004       | ScottDomains/Domain.lean                     | IsAlgebraic, Domain (with the paper's countable-basis condition), BoundedComplete, 8 theorems, Domain Prop                                                                                                                 |
| 3 | r0005       | ScottDomains/Powerset.lean                   | the paper's P N (p. 9): compacts of Set X are exactly the finite subsets, hence Domain (Set ℕ) — the nondegenerate witness                                                                                                 |
| 4 | r0006–r0007 | ScottDomains/ScottHom.lean                   | the continuous function space $D \rightarrow E$ : ScottHom, the pointwise order, CompletePartialOrder, and BoundedComplete when E is — <b>Theorem 7's first sentence in full</b>                                           |
| 5 | r0008       | ScottDomains/StepFunction.lean               | the single step function step k e: continuity (from k compact), the adjunction $\text{step } k \ e \leq f \leftrightarrow e \leq f \ k$ , and compactness in $D \rightarrow E$ (from e compact)                            |
| 6 | r0009       | ScottDomains/FunctionSpaceDomain.lean        | <b>Domain is algebraic</b> — the paper's “they form a basis for $D \rightarrow E$ ”; the two halves of IsAlgebraic use disjoint hypotheses (directedness needs only E bounded complete; the lub needs only D, E algebraic) |
| 7 | r0010       | ScottDomains/CompactFunctionSpace.lean       | every compact function is a <b>finite</b> join of step functions — the finiteness half of the basis claim                                                                                                                  |
| 8 | r0011       | ScottDomains/FunctionSpaceCompactDomain.lean | <b>K(D) is countable</b> , hence Domain (ScottHom $\alpha \beta$ ) — <b>Theorem 7 complete</b>                                                                                                                             |

Next: the §3.1 results that need new definitions first — embedding–projection pairs and (finitary) projections, for Lemmas 4 and 5 and Theorem 6.

## Work counts

- **Reuse from Mathlib — no work (12):** poset, directed, cpo,  $\perp$ , monotone, continuous, Fixed-Point Theorem (1) & operator, Schröder–Bernstein (2), algebraic lattice, product,  $\lambda$ -notation.
- **Generalize / adapt (4):** compact element (IsCompactElement — its *definition* is already stated at [PartialOrder  $\alpha$ ] and so applies to a dcpo verbatim; what is CompleteLattice-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (WithBot/Sum), ideal completion (Order.Ideal).
- **Definitions to define — new ( $\approx 13$ ; 4 done in r0003–r0004, 9 remaining):** way-below  $\ll$  (**done** — ScottDomains/WayBelow.lean), algebraic cpo, **domain**, bounded-complete (**all three done** — ScottDomains/Domain.lean), embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and  $D^\infty$ .
- **Theorems to prove (28):** 28 of the paper's 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

**Bottom line:  $\approx 13$  definitions to define + 28 results to prove**, on top of 12 reused Mathlib foundations. After r0004: **9 definitions + 28 results** remain.

**Lean column legend:** ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize) · ✕ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render  $\rightarrow$  as !, drop fi ligatures, mangle math), so read them as a guide to *what* each result is. Notation:  $\sqsubseteq$  order,  $\sqcup$  directed sup,  $\perp$  bottom,  $\ll$  way-below,  $K(D)$  compacts,  $Fp(D)$  finitary projections.

## §2 Recursive definitions of functions

| §   | Kind | Ref | Concept / statement                                                                                                 | In Lean / Mathlib?                                         |
|-----|------|-----|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 2.1 | Def  | —   | <b>poset</b> (partially ordered set)                                                                                | ✓ PartialOrder                                             |
| 2.1 | Def  | —   | <b>directed</b> subset $M$ : every finite $u \subseteq M$ has an upper bound in $M$                                 | ✓ DirectedOn / IsDirected                                  |
| 2.1 | Def  | —   | <b>cpo</b> : poset in which every directed $M$ has a lub $\bigsqcup M$                                              | ✓ CompletePartialOrder (chains: OmegaCompletePartialOrder) |
| 2.1 | Def  | —   | <b>bottom</b> $\perp$ (least element)                                                                               | ✓ OrderBot                                                 |
| 2.1 | Def  | —   | <b>monotone</b> function                                                                                            | ✓ Monotone / OrderHom                                      |
| 2.1 | Def  | —   | <b>continuous</b> : monotone, $f(\bigsqcup M) = \bigsqcup f(M)$ for directed $M$                                    | ✓ ScottContinuous                                          |
| 2.1 | Thm  | 1   | <b>Fixed-Point Theorem</b> : $f : D \rightarrow D$ continuous $\implies$ least fixed point $\bigsqcup_n f^n(\perp)$ | ✓ OrderHom.lfp (Kleene: OmegaCompletePartialOrder)         |
| 2.2 | Thm  | 2   | <b>Schröder–Bernstein</b> for sets                                                                                  | ✓<br>Function.Embedding.schroederBernstein                 |
| 2.3 | Def  | —   | <b>fixed-point operator</b> (uniform)                                                                               | ✓ OrderHom.lfp / LawfulFix                                 |
| 2.3 | Thm  | 3   | The standard operator is the <b>unique uniform</b> fixed-point operator                                             | ✗ prove                                                    |

## §3 Effectively presented domains

| §   | Kind | Ref | Concept / statement                                                                                                                                                                                                                   | In Lean / Mathlib?                                                                                         |
|-----|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 3.1 | Def  | —   | <b>compact element</b> $x$ : $x \sqsubseteq \bigsqcup M (\text{dir.}) \implies x \sqsubseteq y$ some $y \in M$ ; $K(D)$                                                                                                               | ~ IsCompactElement — def is [PartialOrder], so reusable on a dcpo; its lemmas are all CompleteLattice-only |
| 3.1 | Def  | —   | <b>way-below</b> $\ll$ (approximation relation behind compactness)                                                                                                                                                                    | ✓ ScottDomains.WayBelow (r0003) — $x \ll x \leftrightarrow \text{IsCompactElement } x$ by Iff.rfl          |
| 3.1 | Def  | —   | <b>algebraic</b> cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)                                                                                                                                                   | ✓ ScottDomains.IsAlgebraic (r0004)                                                                         |
| 3.1 | Def  | —   | <b>domain</b> = algebraic cpo <b>whose basis <math>K(D)</math> is countable</b> (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)                                             | ✓ ScottDomains.Domain (r0004)                                                                              |
| 3.1 | Def  | —   | <b>bounded complete</b> : $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i> | ✓ ScottDomains.BoundedComplete (r0004); the compound is [Domain $\alpha$ ] [BoundedComplete $\alpha$ ]     |
| 3.1 | Def  | —   | <b>(countably based) algebraic lattice</b>                                                                                                                                                                                            | ✓ IsCompactlyGenerated (+CompleteLattice)                                                                  |
| 3.1 | Def  | —   | <b>embedding–projection pair</b> $(g, f)$                                                                                                                                                                                             | ✗ define                                                                                                   |
| 3.1 | Def  | —   | <b>projection</b> ; <b>finitary projection</b> $p$ : $p \circ p = p \sqsubseteq \text{id}$ , $\text{im}(p)$ a domain                                                                                                                  | ✗ define                                                                                                   |
| 3.1 | Def  | —   | <b>normal subposet</b> / substructure                                                                                                                                                                                                 | ✗ define                                                                                                   |
| 3.1 | Lem  | 4   | $\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least                                                                                                                                                | ✗ prove                                                                                                    |
| 3.1 | Lem  | 5   | $p$ finitary projection $\implies$ compacts of $\text{im}(p)$ are $\{p(x) : x \in K(D)\}$                                                                                                                                             | ✗ prove                                                                                                    |

| §   | Kind | Ref | Concept / statement                                                                                   | In Lean / Mathlib?                                                                                                          |
|-----|------|-----|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 3.1 | Thm  | 6   | Isomorphism: normal substructures $\cong$ $\text{Fp}(D)$ (finitary projections)                       | ✗ prove                                                                                                                     |
| 3.1 | Thm  | 7   | $D, E$ bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain                    | ✓ <b>proved</b> (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; $D$ bounded complete is not needed |
| 3.2 | Def  | —   | <b>effective presentation</b> $d : \mathbb{N} \rightarrow K(D)$ ; <b>effectively presented domain</b> | ✗ define                                                                                                                    |

## §4 Operators and functions

| §   | Kind | Ref | Concept / statement                                                                           | In Lean / Mathlib?                                             |
|-----|------|-----|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 4.1 | Def  | —   | <b>product</b> $D \times E$                                                                   | ✓ <code>Prod</code> (order/cpo instances)                      |
| 4.2 | Def  | —   | <b>Church’s <math>\lambda</math>-notation</b> (continuous abstraction)                        | ✓ <code>OrderHom / <math>\omega</math>CPO ContinuousHom</code> |
| 4.3 | Def  | —   | <b>smash product</b> $D \otimes E$                                                            | ✗ define                                                       |
| 4.4 | Def  | —   | <b>sum</b> $D + E$ ; <b>lift</b> $D \perp$                                                    | ~ <code>Sum / WithBot, Part</code> (partial)                   |
| 4.x | Lem  | 8   | Currying/iso laws for function spaces over $D, E, F$                                          | ✗ prove                                                        |
| 4.x | Lem  | 9   | Product/function-space iso laws over $D, E, F$                                                | ✗ prove                                                        |
| 4.5 | Lem  | 10  | $D, E$ bounded complete $\implies \rightarrow, \times, \otimes, +, () \perp$ bounded complete | ✗ prove                                                        |
| 4.5 | Thm  | 11  | <b>Ideal completion</b> of a countable pre-order is a domain (all domains so arise)           | ~ <code>Order.Ideal</code> exists $\rightarrow$ prove          |
| 4.5 | Thm  | 12  | Initiality of a continuous algebra satisfying axioms $T$                                      | ✗ prove                                                        |
| 4.5 | Lem  | 13  | $D$ bounded complete $\implies$ powerdomains $D], D[$ bounded complete                        | ✗ prove                                                        |
| 4.5 | Thm  | 14  | Equivalent characterizations of an (algebraic/BC) domain                                      | ✗ prove                                                        |

## §5 Powerdomains

| §   | Kind | Ref | Concept / statement                                                | In Lean / Mathlib? |
|-----|------|-----|--------------------------------------------------------------------|--------------------|
| 5.1 | Def  | —   | <b>powerdomain</b> (non-deterministic outcomes)                    | ✗ define           |
| 5.2 | Def  | —   | <b>Hoare (lower), Smyth (upper), Plotkin (convex)</b> powerdomains | ✗ define           |
| 5.3 | —    | —   | Universal & closure properties (see Lem 13, 28, 30)                | ✗ prove            |

## §6 Bifinite (SFP) domains

| §   | Kind | Ref | Concept / statement                                                            | In Lean / Mathlib? |
|-----|------|-----|--------------------------------------------------------------------------------|--------------------|
| 6.1 | Def  | —   | <b>Plotkin order / bifinite (SFP) domain:</b> bilimit of finite pointed posets | ✗ define           |
| 6.1 | Prop | 15  | Every bounded-complete domain is bifinite                                      | ✗ prove            |
| 6.1 | Thm  | 16  | $D$ bifinite $\implies \text{Fp}(D)$ is an algebraic lattice                   | ✗ prove            |

| §   | Kind | Ref | Concept / statement                                                                                    | In Lean / Mathlib? |
|-----|------|-----|--------------------------------------------------------------------------------------------------------|--------------------|
| 6.2 | Lem  | 17  | $D, E$ bifinite $\implies \rightarrow, \times, \otimes, +, ()^\perp$ bifinite (incl. function space)   | ✗ prove            |
| 6.2 | Thm  | 18  | If $D$ and $D \rightarrow D$ are domains, then $D$ is bifinite                                         | ✗ prove            |
| 6.2 | Lem  | 19  | closure $r : D \rightarrow D$ ( $r \circ r = r \sqcup \text{id}$ ) $\implies \text{im}(r)$ is a domain | ✗ prove            |
| 6.2 | Lem  | 20  | $D$ domain $\implies \text{Fc}(D)$ (finitary closures) is a cpo                                        | ✗ prove            |

## §7 Recursive definitions of domains (universal domain, $D_\infty$ )

| § | Kind | Ref | Concept / statement                                                                                         | In Lean / Mathlib?                                   |
|---|------|-----|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 7 | Def  | —   | <b>recursive domain equation; universal domain</b> / $D_\infty$ (inverse limit)                             | ✗ define                                             |
| 7 | Thm  | 21  | $F$ representable over cpo $U \implies$ a domain $D$ with $D \cong F(D)$                                    | ✗ prove                                              |
| 7 | Thm  | 22  | Any countably-based algebraic lattice $L$ : a closure $r : P(\mathbb{N}) \rightarrow L$                     | $\sim \text{IsCompactlyGenerated} \rightarrow$ prove |
| 7 | Lem  | 23  | The function-space operator is representable over $P(\mathbb{N})$                                           | ✗ prove                                              |
| 7 | Lem  | 24  | $U$ cpo; $\times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)                | ✗ prove                                              |
| 7 | Thm  | 25  | $U$ non-trivial domain representing $\times, \rightarrow \implies U$ <b>universal</b>                       | ✗ prove                                              |
| 7 | Thm  | 26  | Any signature $(s_1, \dots, s_n)$ : combinators $F_1, \dots, F_n$ solving the equations                     | ✗ prove                                              |
| 7 | Thm  | 27  | Any bounded-complete $D$ : a projection of the universal domain onto $D$                                    | ✗ prove                                              |
| 7 | Lem  | 28  | Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ representable over $U$                      | ✗ prove                                              |
| 7 | Thm  | 29  | $D$ bifinite $\implies D^+$ bifinite; solving $D \cong D^+$                                                 | ✗ prove                                              |
| 7 | Lem  | 30  | Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ p-representable over universal bifinite $V$ | ✗ prove                                              |

**Tally (matched):** ✓ reuse Mathlib  $\approx$  **12** (poset, directed, cpo,  $\perp$ , monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product,  $\lambda$ -notation) ·  $\sim$  partial  $\approx$  **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · ✗ define / prove  $\approx$  **44**, of which 4 are done ( $\ll$  r0003; algebraic, domain, bounded-complete r0004)  $\rightarrow$  **40 remaining** — the bifinite / powerdomain /  $D_\infty$  development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.